

Verifai

Synthetic Media Detection Engine

Datathon 2026

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The Trust Crisis

Erosion of Authenticity

Deepfakes are evolving faster than human intuition.
Authenticity is no longer a given in a digital-first world.
Most existing verification tools require uploading sensitive personal data to centralized cloud servers.

Operational Barriers

Privacy Tax: Users must sacrifice anonymity for accuracy.
Latency: Server-side checks are slow, prone to downtime, and break the user workflow.

The Privacy Paradox



Cloud Burden

Traditional verification compromises user anonymity for server-side processing power.



Surveillance

Centralized AI probing turns every verification request into a data point for tracking.

Dependency

Constant internet reliance makes verification fragile and vulnerable to API throttling.

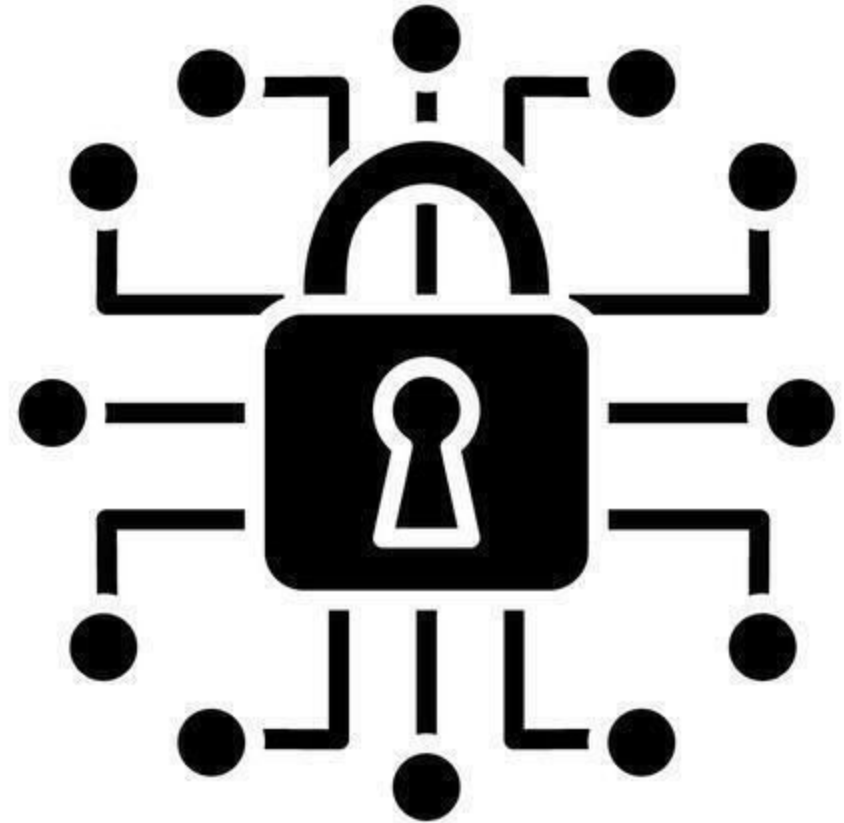
On-Device Intel

Edge Architecture

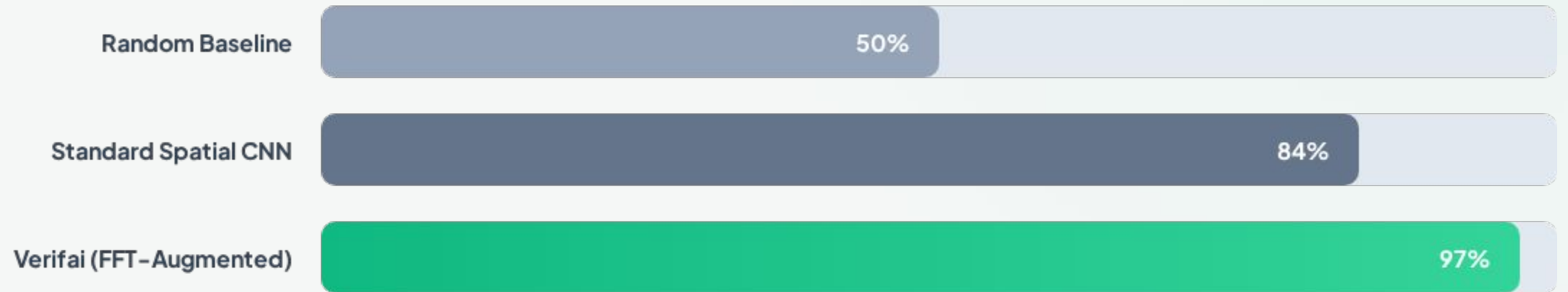
Verifai is a browser-native extension that never lets your raw data leave the device sandbox.

ONNX Web Runtime: Leverages WebAssembly and WebGL for hardware-accelerated local inference.

Offline Capability: Functions entirely in isolation. No server, no surveillance, no data leaks.



Performance Metrics



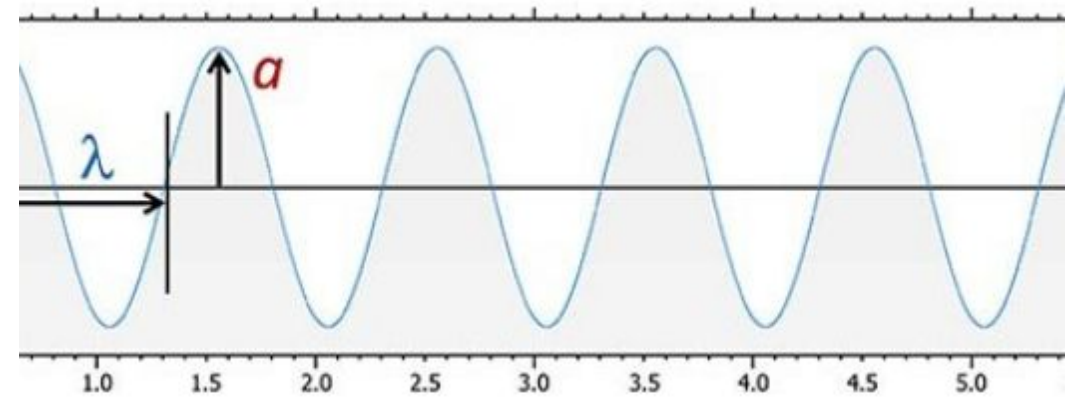
Validated on **CIFAKE (120,000 images)**. Achieving elite accuracy with a sub-3MB model footprint for instant browser loading.

The Hybrid Engine

 Spatial neural network extraction

Spatial Extraction

Standard convolutions identify geometric inconsistencies and micro-artifacts in the pixel domain.



Spectral Gating

Fast Fourier Transforms (FFT) isolate periodic synthetic artifacts invisible to spatial filters.

The Threshold Strategy

0.5
Decision Boundary

No Binary Classification

We reject rigid binary labels that mask model uncertainty. Verifai surfaces raw confidence scores to invite "human-in-the-loop" verification.

Buckets: 0–0.25 (Fake), 0.26–0.75 (Unsure), 0.76–1.0 (Real). By flagging ambiguity, we prevent false confidence.

Technical Guardrails



Pydantic Integrity: Strict schema validation and programmatically clamped output ranges prevent downstream edge-case failures.



SSRF Protection: Hostnames are resolved and blacklisted for private/loopback ranges to prevent internal infrastructure probing.

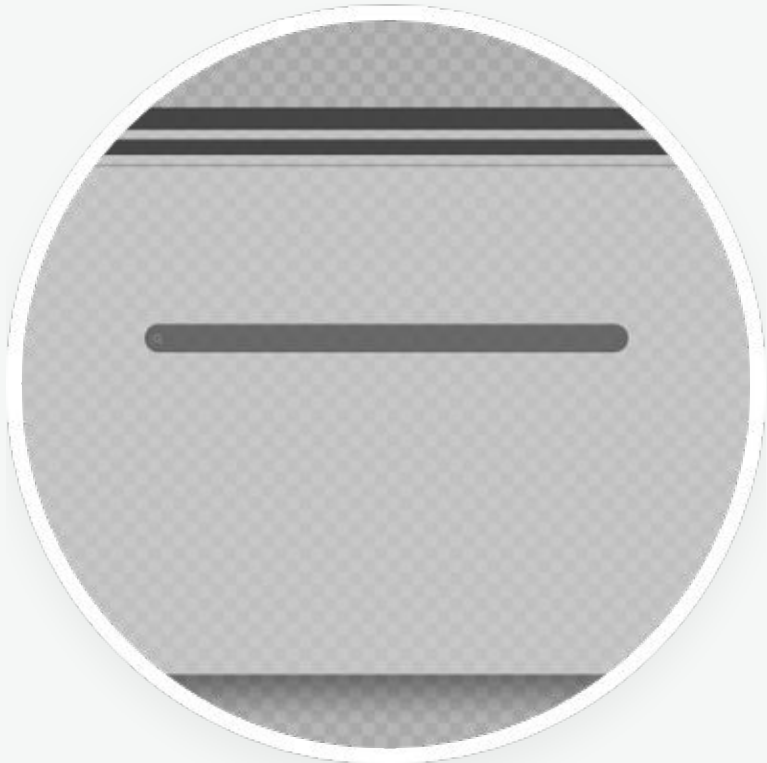


Resource Throttling: Strict 5MB file caps and MIME-type enforcement mitigate payload-bloat and Denial-of-Service vectors.



Reproducibility: Fixed random seeds and deterministic kernel execution ensure stable model weights and verifiable results.

Built for the User



Instant & Invisible

Integration into daily workflows without friction or data leaks.

Designed as an augmentation, not a replacement for human judgment.

Lightweight: Tiny model footprint ensures the extension doesn't slow down the browser.

Seamless: Right-click context menu integration for instant media provenance checks.

Demo

<https://youtu.be/FF4WazmO36U>

Questions?

Honest AI. Local Privacy. Digital Truth.

github.com/stephkmn/datathon26

Closing the Gap Between SWE and AI | Datathon

2026

Image Sources



https://static.vecteezy.com/system/resources/previews/057/645/473/non_2x/digital-network-security-icon-cyber-protection-symbol-with-a-padlock-and-circuit-connections-encryption-privacy-and-data-safety-in-online-systems-cybersecurity-and-secure-access-concept-vector.jpg

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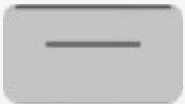
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